

## Power series convergence - 6.1

Example: For which  $x$  does  $\sum_{n=0}^{\infty} n(x-2)^n$  converge?

Ratio test  $\lim_{n \rightarrow \infty} \left| \frac{(n+1)(x-2)^{n+1}}{n(x-2)^n} \right|$

$$= \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} (x-2) \right| = |x-2| \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \right| = |x-2|$$

If  $|x-2| < 1$ , then  $\sum_{n=0}^{\infty} n(x-2)^n$  converges absolutely

and if  $|x-2| > 1$ , then it diverges.

If  $|x-2| = 1$ , then the ratio test is inconclusive.

To say that  $|k-2| < 1 \Rightarrow$  abs. convergence  
to say that we have abs. convergence  
for  $k \in (1, 3)$ . We have divergence for  
 $k < 1$  and  $k > 3$ . What happens for  $k=1, 3$ ?

$$\sum_{n=0}^{\infty} n(k-2)^n \text{ at } k=1 \text{ is } \sum_{n=0}^{\infty} n(-1)^n,$$

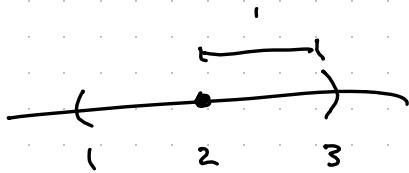
and  $\lim_{n \rightarrow \infty} n(-1)^n$  DNE, so we have divergence  
at  $k=1$  by the divergence test.

$$\sum_{n=0}^{\infty} n(k-2)^n \text{ at } k=3 \text{ is } \sum_{n=0}^{\infty} n, \text{ which also}$$

diverges by the divergence test.

We have shown that  $\sum_{n=0}^{\infty} n(x-2)^n$  converges if and only if  $x \in (1, 3)$ .

We call this the interval of convergence,



and we call 1 the radius of convergence.

Theorem: For a power series  $\sum_{n=0}^{\infty} c_n(x-a)^n$ , exactly one of the following is true:

- i. We have convergence for all  $x$ .
- ii. We have convergence only at  $x=a$ .
- iii. There's a number  $R$  such that we have convergence if  $|x-a| < R$  and divergence if  $|x-a| > R$ . (It may converge or diverge if  $|x-a| = R$ .)

Example: Find the radius and interval of convergence of  $\sum_{n=1}^{\infty} \frac{1}{n} (x-4)^n$ .

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{1}{n+1} (x-4)^{n+1}}{\frac{1}{n} (x-4)^n} \right| = \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} (x-4) \right| = |x-4|$$

We have convergence for  $|x-4| < 1$

divergence for  $|x-4| > 1$ , so the

radius of convergence is  $R=1$ .

The interval of convergence contains  $(3, 5)$ , but we need to check the endpoints.

$$x=3: \sum_{n=1}^{\infty} \frac{1}{n} (-1)^n$$

converges by  
the AST

$$x=5: \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges by  
p-test ( $p=1$ )

Interval of  
convergence is  
therefore  $[3, 5)$ .

Proposition: Suppose  $\sum c_n x^n$  and  $\sum d_n x^n$  converge to  $f(x)$  and  $g(x)$  on some interval  $I$ .

- i.  $\sum (c_n \pm d_n) x^n$  converges to  $f(x) \pm g(x)$  on  $I$
- ii. For any  $m \geq 0$  and  $b$  a real number,  $\sum b c_n x^{n+m}$  converges to  $b x^m f(x)$  on  $I$
- iii.  $\sum c_n (b x^m)^n$  converges to  $f(b x^m)$  for all  $x$  such that  $b x^m \in I$ .

Example:  $\sum_{n=0}^{\infty} x^n$  converges to  $\frac{1}{1-x}$  on  $I = (-1, 1)$ .

By the above,  $\sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n x^n$  converges to  $\frac{1}{1-(-x)} = \frac{1}{1+x}$  on  $(-1, 1)$ .

$$\sum_{n=0}^{\infty} 2^n x^n = \sum_{n=0}^{\infty} (2x)^n \quad \text{converges to}$$

$$\frac{1}{1-2x} \quad \text{for all } x \text{ such that } 2x \in (-1, 1),$$

$$\text{i.e. } -1 < 2x < 1 \iff -\frac{1}{2} < x < \frac{1}{2}, \text{ so}$$

$$\sum_{n=0}^{\infty} 2^n x^n \quad \text{converges to} \quad \frac{1}{1-2x} \quad \text{on } I = \left(-\frac{1}{2}, \frac{1}{2}\right).$$